

DiSandbox: A Low-cost Digital Sandbox Tool to Support Psychological Analysis and Therapy for Left-behind Children

Yan Shi
hdushiyan@163.com
Hangzhou Dianzi university
China

Yiwen Lu
luyiwen@hdu.edu.cn
Hangzhou Dianzi university
China

Lidan Gong
gonglidan@hdu.edu.cn
Hangzhou Dianzi university
China

Lijuan Liu*
liulijuan@hdu.edu.cn
Hangzhou Dianzi university
China

ABSTRACT

This paper introduces DiSandbox, a low-cost digital tool designed to support left-behind children in rural China facing psychological challenges. Comprising a range of miniature molds within a sand tray, children can freely express themselves, with an inbuilt camera capturing their creations. These images are sent to their parents' mobile application, where an AI assistant interprets the works and assesses the children's psychological conditions. The platform also offers professional psychological counseling, delivering personalized support. Our research shows that DiSandbox can effectively analyze children's creations to provide initial psychological assessments, encouraging parents to take greater interest in their children's mental health.

KEYWORDS

Sandbox game, Psychological assessment tools, Digital design, Toolkits, Left-behind children

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1 INTRODUCTION

Left-behind children are those who are forced to stay in their registered residence while their parents migrate to other areas and are unable to live with them [10]. In 2015, UNICEF estimated that there were approximately 68.77 million children left in rural areas of China while their parents migrated for work, with some provinces having as high as 40 percent of rural children being left behind

[11]. Long-term separation from parents leads to limited communication between left-behind children and their parents about their emotional state, which can result in psychological, behavioral, and personality deviations. The emotional distance between parent and child can cause various physical and psychological issues [13], particularly learning anxiety, self-blame tendencies, and physical symptoms [15]. For most left-behind children, seeking professional psychological counseling or therapy requires traveling to urban areas or even farther, which is a significant financial burden. Therefore, in this project, we focus on low-cost methods for psychological assessment and healing for left-behind children.

Sandbox therapy, a psychotherapy method applicable to all ages, effectively prevents emotional problems, promotes self-insight, and addresses maladaptive environments [23]. Among all auxiliary tools in psychotherapy, sandbox is particularly suitable for children and their migrant parents [3]. It provides a safe space for them to play and create without direct adult involvement. The tangible nature of the sandbox allows them to replace and express emotions through objects. Additionally, sandbox therapy helps children develop problem-solving and decision-making skills to cope with real-life challenges. Despite its numerous advantages, sandbox therapy also has limitations. Firstly, it requires therapists with extensive knowledge to accurately interpret and guide children's sandbox activities. Secondly, it is mostly used in professional treatment settings or psychological consultation rooms that charge per session. This can be financially burdensome for left-behind children who may require multiple sessions to truly address their emotional issues.

In this article, we propose a small and cost-effective family sandbox as a communication intermediary between left-behind children and their parents. This innovative solution aims to conveniently achieve psychological education or therapy for left-behind children and make it easier for parents to communicate with their children in real time. DiSandbox is a digital sandbox tool with 28 different types of molds, allowing children to express themselves naturally and freely during the game process. An integrated camera records the children's creations, while artificial intelligence technology analyzes their works to provide preliminary evaluation. In addition, DiSandbox also enables parents to observe and understand the children's mental health status, and timely detect and address potential problems. Furthermore, DiSandbox provides families in need of help with the option to choose a professional psychologist for diagnosis, greatly facilitating access to professional psychological services. In

*Corresponding author

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summary, DiSandbox is an interactive and innovative digital tool designed for psychological assessment and treatment purposes, and will positively impact the mental health of left-behind children. The contributions of this paper are as follows:

1. AI-powered digital sandbox tool for psychological assessment and therapy of left-behind.
2. A communication platform was established to support the mental health of left-behind children in rural areas, connecting them with their parents and psychologists.
3. A user study evaluated the effectiveness of DiSandbox tasks and children's experience using it.

2 RELATED WORK

2.1 Design for the psychological well-being of left-behind child

"Left-behind life" refers to the situation where children are separated from one or both parents during childhood or adolescence and live with non-parental guardians for over 6 months [5]. This phenomenon not only negatively impacts the physical and mental health of left-behind children but also hinders their psychological resilience, self-esteem, and emotional well-being due to prolonged parental separation. They experience heightened feelings of loneliness, depression, anxiety [2], as well as encounter difficulties in learning, forming interpersonal relationships, and managing emotions. Consequently, researchers are increasingly focusing on strategies to address the mental health issues faced by left-behind children. For instance, Zeng et al. (2023) [24] developed wearable soft robots that facilitate timely non-verbal interaction between left-behind children and parents to strengthen emotional connections. By touching the robot, children can convey their emotions to their parents who receive tactile signals on their wrists and provide prompt care and comfort. Additionally, Lu's team (2020) [14] discovered that enhancing communication initiative between left-behind children and their parents can improve these children's psychological adaptability; thus they created an online communication program aimed at establishing contact. However, while these designs have alleviated the psychological problems of left-behind children to some extent, there are still some limitations in existing research. Most designs mainly focus on providing companionship and emotional support, but their functions in dealing with more complex psychological problems such as stress and anxiety are relatively inadequate.

2.2 Sandbox Therapy

Sandbox therapy is a nonverbal psychotherapy method that involves players using sand and sand toys to arrange their environment and engage in a dialogue with their inner conscious and unconscious minds based on positive imagination, thereby achieving traumatic recovery and personality improvement [1]. Compared with other psychotherapy techniques, sandbox therapy has a natural advantage in the treatment of children's psychological problems [22]. In sandbox, children can gain more emotional experiences and externalize these experiences into the sand box layout, which reflects their emotions and psychological states. Unnsteinsdottir Krist's research has shown that sandbox can improve children's learning ability and emotional disposition to a certain extent [20].

Today, sandbox therapy has been successfully applied to the treatment of various psychological disorders in adults and children, such as traumatic stress, disability, language problems, etc. In addition to the traditional sandbox mode, the integration of virtual reality technology into sandbox has also brought new possibilities for psychotherapy [21]. Despite its many advantages, the treatment of left-behind children in rural areas requires professional equipment and guidance personnel, and the high cost is not suitable for psychological assessment and rehabilitation.

2.3 Artificial Intelligence-Assisted Psychological Assessment and Therapy

In recent years, the widespread application of artificial intelligence (AI) in psychological assessment and treatment has gained considerable attention. Numerous studies have made significant progress in recognizing psychological symptoms, assessing risks, and providing personalized treatment recommendations. For example, scholars have explored AI-based systems that analyze facial expressions and voices to identify users' emotions [17]. Additionally, there are studies that use text analysis on social media platforms to detect potential mental disorders among users [12]. In terms of risk assessment, research has developed machine learning algorithms for evaluating suicide risk in adolescents and predicting future mental disorders by analyzing their medical records through big data analysis [8]. Furthermore, AI has played an active role in offering personalized treatment recommendations with the development of online psychological counseling robots that provide tailored suggestions based on users' specific issues [6]. However, this field still faces several challenges including improving AI accuracy and reliability, ensuring user privacy during data collection and analysis processes, as well as designing and implementing effective AI interventions for promoting users' mental health. Further research is needed to explore potential limitations and strategies to overcome them while conducting more empirical studies to gain a better understanding of AI effectiveness across different contexts and populations.

3 FORMATIVE RESEARCH

To comprehend the mental health status of left-behind children and ascertain sandbox game component design requirements, we conducted a questionnaire survey on 115 school children in rural areas of Sichuan Province, China. Additionally, semi-structured interviews were carried out with 10 parents, primarily focusing on family environment, parent-child relationship, and cognition of sandbox games. Interviews and questionnaires were administered after obtaining informed consent forms to ensure participant privacy protection. Findings from both revealed a significant prevalence of left-behind children in the region, with 70 percent classified as such. Approximately one-third exhibited various mental health issues related to learning anxiety, loneliness tendencies, hostility traits among others based on evaluations using a mental health diagnostic test scale. Comprehensive analysis and insights derived from this study's outcomes led to three key design inspirations:

1. Development of simple yet accessible psychological assessment tools and therapeutic interventions.
2. Facilitating timely parental understanding regarding their child's psychological state.



Figure 1: (a)The hardware part of DiSandbox system overview. (b) All types of sand toys. (c) A child is creating a sandbox scene.

3.Exploring low-cost methods for psychological analysis and therapy.

4 DISANDBOX DESIGN

Based on the design inspirations mentioned above, we developed the DiSandbox system, which includes both the hardware component of the sandbox model and the software analysis application.

4.1 Hardware Design

As shown in (Figure.1(a)), the dimensions of the DiSandbox sandbox are consistent with those of a professional portable sandbox in a counseling room, measuring 46.5cm×36.5cm×5.5cm. Considering cost-effectiveness, we used wooden materials for construction, and the sand grains used are of the same specifications as those used in professional sandboxes, with a cost of approximately 15 dollars. The sandbox accessories include seven categories: people, animals, plants, buildings, vehicles, fences, and natural objects [9, 16], totaling 25 pieces (Figure.1(b)). An ESP32-CAM camera module is placed on the side to record the children's creative process, with data transmission and reception via Wi-Fi. During the creation process, children are free to arrange the accessories and have no strict requirements regarding the creation theme or duration (most creators can finish the sandbox operation within 15-30 minutes). The results of the creation can be recorded and transmitted to the software side.

4.2 Software Design

Once the parents, who are far away, connect to the sandbox through the mobile application, they can view their child's creation results (Figure.2(a)). Parents can choose to use the AI assistant for artwork analysis, which provides an initial assessment of the child's psychological state (Figure.2(b)). By clicking on each accessory, parents can gain a more detailed understanding of the information and symbolic meaning behind their child's sandbox creation. For example, in sandbox therapy, dinosaurs symbolize unconscious forces. If a child places dinosaurs in their sandbox, it may indicate feelings of fear, anxiety, loneliness, and inner conflicts, but also a desire to seek help and overcome difficulties, with a primary focus on positive energy [19] (Figure.2(c)).

4.3 AI-Assisted sandbox Creation Analysis

When the sandbox pictures are uploaded to the parents' end, the system will conduct a comprehensive analysis of various aspects

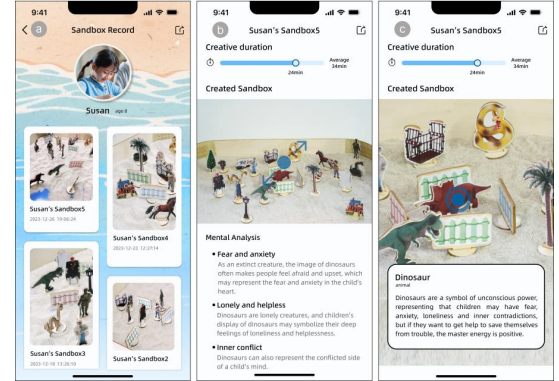


Figure 2: The DiSandbox app design. (a) Parents can view their children's creations. (b) Check out the sandbox created by children. (c) Enlarge the picture, click to interact, parents can get related interpretation of sand toys.

including production time, sand tool selection, and placement position [25]. We utilized Teachable Machine's sand tool classification model by training it with a large number of labeled sand tool images and optimizing it through multiple iterations to enhance its ability in detecting different categories of sand tools [4, 18]. In DiSandbox, we exported and integrated this trained model into the application. Based on image recognition results, we analyze the placement position and category of sand tools, providing parents with interpretations of sand tool labels as well as basic psychological analysis of children.

4.4 Personalized Expert Counseling Services for Optimized Experience

When evaluating sandbox works, analysts should consider not only the image's overall content but also the creator's background, psychology, culture, symbolism and other knowledge to complete a thorough reasoning process[7]. To provide a more comprehensive analysis of left-behind children's psychological status and ensure professionalism and effectiveness of intelligent sandbox applications, we offer expert consultation services for parents to choose from based on AI preliminary evaluation results.

4.5 Privacy and Security

The intelligent sandbox system ensures privacy and safety for left-behind children by allowing only matched parents to view their

uploaded pictures, while the selection of psychological doctor services requires parental approval.

5 USER STUDY

5.1 Participants and Procedure

To assess the usability and user experience of DiSandbox, we recruited a sample of 7 left-behind families comprising 7 parents and 7 children. The average age of the children was 12.29 years old ($SD=\pm 1.83$). They evaluated the tool for one month, focusing on exploring the following research questions:

- Can the tool effectively support children in creating sandboxes and providing an initial mental health assessment?
- Does the tool facilitate parental attention towards their children's mental health?
- What are the experiences reported by children while using this tool?

Before the experiment, we explained its purpose to all participants and obtained their informed consent through signed documents. A qualified psychologist conducted psychological assessments on all children involved prior to starting the experiment. The results showed that 4 out of 7 children had mild psychological issues, while 3 displayed mental health concerns. Over the next month, the children engaged in sandbox creation activities facilitated by DiSandbox. To ensure comparability among participants' data, sandbox creation sessions were scheduled once a week with predetermined themes (refer to Table 1). The children created their sandboxes in a consistent and nurturing environment, with video recordings capturing both the process and outcomes of their creations. These finished works were shared with parents who could review them alongside an assessment of their child's psychological state provided by an AI assistant.

After the one-month testing period, we invited the psychologist to reevaluate each child's psychological state. Additionally, semi-structured interviews were individually conducted between each child and their respective parents. Two professional psychologists were also invited to analyze video data depicting the children's creations and provide psychological interpretations for each individual.

Table 1: Four weeks of creative themes

Time of Creation	Theme of Creation
1st week	Energy
2nd week	Travel
3rd week	Link
4th week	Birth

5.2 Result

5.2.1 Task Effectiveness.

Four sandbox healing works of 7 children were analyzed, along with the corresponding preliminary psychological state assessments provided by AI (Figure.3(a-d)). Experts assessed the accuracy of both AI interpretation results and psychological state assessment results for a total of 28 works using a 5-point Likert scale, where higher scores indicated higher accuracy. DiSandbox's interpretation of

sandbox creation works showed slightly higher average accuracy ($M = 4.14$, $SD=0.86$) compared to the preliminary psychological state assessment results ($M = 3.96$, $SD=0.81$), with a significant difference observed ($p = .045^{***}$)(figure.4). The complexity of human emotions and behavior involved in psychological assessments may be a reason for the limitations of current AI technology in this area. However, AI assistants can make accurate interpretations of sandbox works by analyzing details such as the shape, color, and arrangement of objects. With the help of big data analysis, AI assistants are able to identify and understand the rich information contained in these details, thereby providing effective interpretations.

We invited psychological doctors to re-evaluate the mental health of 7 left-behind children after a month-long experiment. The results showed that 3 children maintained a healthy mental state, while 2 out of 4 children with mild mental health problems improved and became mentally healthy. However, the other 2 still had mild mental health issues. Despite this, their assessment scores improved compared to before the experiment and they displayed positive behavior during the experiment. This one-month experiment provided preliminary evidence for the functional effectiveness of DiSandbox and its positive impact on the mental health of left-behind children. In future studies, we will conduct long-term experiments to further evaluate the effectiveness of DiSandbox.

5.2.2 Parents' concern for their children's mental health.

During the interview, we discovered a significant increase in parents' concern for their children's psychological well-being. They not only monitor their children's sandbox creations through DiSandbox and read relevant analyses but also inquire about their moods and feelings more frequently during daily phone calls or WeChat exchanges. One mother mentioned, "We now pay closer attention to our children's inner world. I often ask them about their experiences and emotions at school, specifically checking if they have encountered any instances of bullying."

5.2.3 The User Experience Design for Children.

The children loved DiSandbox, seeing it as a sandbox for free expression and creation rather than just a psychotherapy tool. "I love this sandbox! It lets me create my favorite Jurassic World." "I wish there were more sand tools so I can express scenes that I have in mind." "I love the camera feature too! I just talked to my mom on the phone about what I made!" Although they didn't mention its impact on their mental health, they are eager to continue playing with DiSandbox and share their creations with their parents.

6 LIMITATIONS AND FUTURE WORK

Based on the analysis and healing of left-behind children's psychological status, we developed DiSandbox, a cost-effective digital sandbox tool that supports remote interaction. The system offers convenient and affordable physical and mental healing for left-behind children, while also serving as a communication medium between parents and children to enhance their relationship. In future work, we aim to introduce more diverse modes to promote communication between left-behind children and parents, addressing the root causes of frequent psychological problems in these children.



Figure 3: The children created sandbox works. Figures a-d are respectively about the themes of energy, travel, connection and birth.

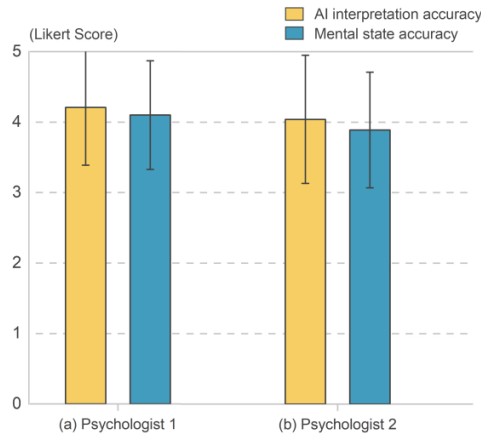


Figure 4: Two psychologists rated the accuracy of the DiSandbox tool in sandbox work interpretation and in preliminary assessment of psychological states.

7 CONCLUSION

In this paper, we propose DiSandbox, a cost-effective digital toolbox designed to address the mental health issues of left-behind children. By integrating artificial intelligence technology, DiSandbox offers an interactive and innovative digital psychological assessment and healing method through recording, analysis, consultation, and archiving. Our research demonstrates that DiSandbox can analyze children's sandbox works to provide preliminary psychological assessment results with recognized accuracy by experts in psychology. Interviews with children and parents reveal that children perceive DiSandbox as a toy and willingly engage in interactive play. Parents appreciate its intuitive understanding of their children's mental health status, reasonable price point, and additional channels for psychological consultation. Overall, the digital format of DiSandbox presents a valuable tool for enhancing the mental well-being of left-behind children.

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